National University of Computer & Emerging Sciences, Karachi
Spring-2022 CS-Department
Mid-term II
5th April, 2023; 11:30 am to 12:30 pm

Course Code: SS 2007	**Course Name: Technical & Business Writing**
Instructor Name:Ms. Nazia Imam, Ms. Javeria Ali, Ms. Madiha Rehman, Ms. Sabeen Amjad	
Student Roll No:	**Section No:**

Instructions:

- Return the question paper.
- Read each question completely before answering it. There are **2 questions and 2 pages.**
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- ANSWER ALL QUESTIONS IN THE ANSWER COPY. QUESTIONS ANSWERED ON THE QUESTION PAPER WILL NOT BE SCORED.
- **The invigilator present is not supposed to tell the meaning of any word.**
- Be precise and to the point.

Time: 1 hour **Max Marks: 30 points**

Question No. 1 CLO:4

Given below is an advertisement for Computer Science Interns. Read it and write the following sections of a Resume targeting this internship position.

1. **Objective [2]**
2. **Technical Competencies/Skills [3]**
3. **Relevant Academic Projects (Only Two) [4]**
4. **Soft Skills sections (only Two) [4]**

Reston is seeking Computer Science interns who possess the ability to thrive on new challenges as an Android developer. Through this internship program, you get the opportunity to work on exciting projects for one or more of our prestigious clients across the globe.
The interested candidates who are currently pursuing bachelor's degree in computer science or related field are encouraged to apply with at least one semester/quarter remaining after internship.
Also, a year or two of experience programming in C++, Java or other computer programming languages. The potential candidates with influential speaking skills for negotiation and conflict management will be preferred.

Question No. 2 CLO:3

a. **Rearrange the following information for the title page of a report in the correct order. [Marks 03]**

1. 2005
2. Supervised by: Dr. Peter Evans
3. Touka Harlot
4. Department of Computing
5. Schema Dependent Peer-to-Peer Data Integration
6. Oxford University

b. Match the correct option with the correct definition. Write only the numbers of the matched pair. [Marks 03]

a. Objectives	1. provides an overview of the main points of a larger report. It is often written to share with individuals who may not have time to review the entire report.
b. Abstract	2. provides the recipient with a specific context in which to place the larger document and simultaneously gives the sender a permanent record of having sent the material.
c. Scope	3. a brief SUMMARY of your work which is capable of being read independently of it.
d. Executive Summary	4. Shortcomings of the paper
e. Letter of Transmittal	5. describes the extent to which the research question will be explored in your study.
f. Limitations	6. describe what your research is trying to achieve and explain why you are pursuing it.

c. Read the following abstracts. Identify and write 2 objective/purpose statements (in your own words) mentioned implicitly or explicitly in the abstracts given. [Marks 06]

1. To ensure the safety of autonomous Multi-rotor UAVs flying in urban airspace, they should be capable of avoiding collisions with unpredictable dynamic obstacles, such as birds. UAVs must consider both relative position and relative velocity to avoid moving obstacles. Model predictive control (MPC) can consider the multiple collision avoidance constraints in a constrained optimization framework. This study proposes a chance-constraints based on obstacle velocity (CCOV) method, which can be combined with previous positional chance constraint methods to account for uncertainty in both position and velocity. This effectively prevents collision with high-velocity obstacles, even in a noisy environment. The proposed method has been performed on a numerical simulation built in MATLAB.

2. Eating is an essential activity of daily living (ADL) for staying healthy and living at home independently. Although numerous assistive devices have been introduced, many people with disabilities are still restricted from independent eating due to the devices' physical or perceptual limitations. In this work, we present a new meal-assistance system and evaluations of this system with people with motor impairments. We also discuss learned lessons and design insights based on the evaluations. The meal-assistance system uses a general-purpose mobile manipulator, a Willow Garage PR2, which has the potential to serve as a versatile form of assistive technology. Our active feeding framework enables the robot to autonomously deliver food to the user's mouth, reducing the need for head movement by the user. The user interface, visually-guided behaviors, and safety tools allow people with severe motor impairments to successfully use the system. We evaluated our system with a total of 10 able-bodied participants and 9 participants with motor impairments. Both groups of participants successfully ate various foods using the system and reported high rates of success for the system's autonomous behaviors. In general, participants who operated the system reported that it was comfortable, safe, and easy-to-use.

d. Read the following background and develop a problem statement for it. (Word count: 35-50)

In recent years, technology has played a significant role in shaping various aspects of our lives, including education. The widespread adoption of online learning platforms, such as Coursera and Udemy, has allowed learners to access high quality educational content from the comfort of their homes. While online learning offers several benefits to learners, it is also associated with some challenges, such as a lack of personal interaction and guidance. To address these challenges educators have started exploring the potential of Artificial Intelligence (AI) in online learning.